

Selective Geometry-Guided Generation for Single-View Scene Reconstruction

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Abstract

Single-view scene reconstruction is dominated by a simple asymmetry: visible regions can be copied from the input view, while disoccluded regions must be hallucinated. Existing feed-forward and generative methods are usually evaluated with full-image metrics, which hide this failure mode. We propose a selective generative refinement method that improves disoccluded regions without damaging visible regions. Given a single input image and a target camera path, we use a feed-forward geometry expert (Flash3D) and a generative reconstruction backbone (Gen3R). During Gen3R denoising, we inject the geometry expert’s evidence *only* in invisible regions, leaving visible regions unchanged. A test-time reliability gate decides whether injection should be enabled for each scene, preventing failures when the baseline is already better than the geometry expert. A mechanism analysis shows that the invisible-region quality gap between the geometry expert and the generative baseline explains injection benefit with correlation $r = 0.985$ on 16 scenes, reproduced at $r = 0.982$ when scaling to 166 scenes. The large-scale study sharpens the claim: injection adds +2.49 dB on hard disoccluded scenes but −2.52 dB on easy scenes where the baseline is already strong, confirming that selective gating is essential. Invisible-region perceptual quality improves (SSIM 0.579 → 0.725, LPIPS 0.0948 → 0.0784) while visible-region quality is provably unchanged.

1 Introduction

Reconstructing a 3D scene from a single image requires solving two very different sub-problems. Pixels that remain visible in a novel view can be warped or copied from the input and are recovered accurately by feed-forward methods. Pixels that become *disoccluded* in the novel view have no source evidence and must be synthesized, which is where feed-forward methods degrade to blurry or blank content. Full-image PSNR averages over both regions and therefore hides this failure: a method can score well while producing empty

disocclusions.

We take the asymmetry seriously and evaluate visible and invisible regions separately. We combine two off-the-shelf systems that we treat as *components*, not competitors: a feed-forward per-pixel Gaussian reconstructor (Flash3D) that supplies sharp geometric evidence, and a generative reconstruction backbone (Gen3R) that hallucinates plausible content. Our method injects the feed-forward evidence into the generative denoising process, but *only* in the invisible region, so that visible pixels are never modified.

Crucially, injection is not universally beneficial. When the generative baseline already reconstructs a disoccluded region well, forcing feed-forward evidence in degrades it. We therefore add a test-time reliability gate that predicts, per scene, whether injection will help. Our contributions are:

1. **Asymmetric geometry injection:** a region-selective mechanism that edits only the invisible latent, leaving visible regions provably untouched (visible-region $\Delta = 0$ dB).
2. **Test-time reliability gate:** a scene-level decision rule that recovers 98% of an oracle gate’s gain and turns a −11.7 dB worst case into −0.17 dB.
3. **Mechanism analysis:** the invisible-region gap between the geometry expert and the generative baseline explains injection benefit at $r = 0.985$, reproduced at $r = 0.982$ on a 10× larger, harder set.
4. **Region-aware evaluation:** separate visible/invisible PSNR, SSIM and LPIPS on 166 RealEstate10K scenes, plus non-parametric bootstrap confidence intervals for every headline number.

2 Related Work

Feed-forward 3D reconstruction. pixelSplat [1], MVSplat [2] and DepthSplat [10] regress 3D Gaussians [4] from sparse views, and Flash3D [8] extends this to a single image. These methods are accurate

in visible regions but produce degenerate content in disocclusions; we use Flash3D as our geometry expert rather than as a baseline.

Generative novel view synthesis. Diffusion-based methods such as ViewCrafter [12], CAT3D [3] and GenWarp [7] synthesize plausible unseen content but can drift from the true geometry. Our backbone (Gen3R) is of this family; we keep its generative prior for hallucination while grounding disocclusions with feed-forward evidence.

Injecting geometry into diffusion models. RePaint [5] conditions diffusion inpainting on known pixels; REPA [11] aligns diffusion features to pretrained representations during *training*; Geometry Forcing [9] makes a video-diffusion world model 3D-aware by aligning to geometric features at training time. Our method differs on three axes: (i) *timing* — we intervene at *test time* rather than during training; (ii) *locality* — we edit only the disoccluded region rather than the whole latent; and (iii) *selectivity* — a reliability gate turns injection off when it would hurt.

Reliability, routing and test-time guidance. Our gate is a form of test-time routing: predict, from observable signals, whether an expert should be trusted for a given input. We show a scene-level gate is a strong first step and that the signal must ultimately be measured at the frame level.

3 Method

3.1 Overview

Given a single input image and a target camera path, Flash3D renders feed-forward evidence for each target view (Fig. 1). We encode these renders into the RGB half of Gen3R’s latent space. We also run Gen3R once to obtain a clean baseline latent. During a second Gen3R denoising pass, we modify only the invisible region of the RGB latent.

3.2 Asymmetric Geometry Injection

Let $M \in \{0, 1\}$ be the visibility mask (1=visible). For each denoising step we replace the latent only where $M = 0$:

$$z' = M \odot z + (1 - M) \odot [(1 - w)z + w z_{\text{f3d}}], \quad (1)$$

where z is the current Gen3R latent, z_{f3d} is the encoded Flash3D evidence, and w is an injection weight. Because the visible region uses $M = 1$, visible pixels are never modified.

Algorithm 1 Selective geometry-guided generation

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1: Input: image  $I$ , target poses  $P_t$ , visibility masks  $M_t$ 
2:  $z_{\text{f3d}} \leftarrow \text{Encode}(\text{Flash3D}(I, P_t))$ 
3:  $\text{base\_lat} \leftarrow G(I, P_t)$   $\triangleright$  one clean baseline denoise pass
4: if not Gate( $\text{f3d}$ ,  $\text{base\_lat}$ ) then
5:   return Decode( $\text{base\_lat}$ )  $\triangleright$  gate rejects: keep baseline
6: end if
7: for each denoising step do
8:    $w \leftarrow \text{InjectionWeight}(z, z_{\text{f3d}}, \text{base\_lat})$ 
9:    $z \leftarrow M_t \odot z + (1 - M_t) \odot [(1 - w)z + w z_{\text{f3d}}]$ 
10: end for
11: return Decode( $z$ )

```

3.3 Clean-Space Confidence

The handcrafted injection weight uses agreement between Flash3D and the clean Gen3R baseline latent, so evidence is trusted more where the two already agree and downweighted where they diverge.

3.4 Learned Injection Weight

We additionally train a small *InjectionWeightNet* that predicts w from local latent statistics. It matches the handcrafted confidence, showing the mechanism can be learned end-to-end rather than hand-tuned.

3.5 Test-Time Reliability Gate

Injection is enabled for a scene only if an observable reliability signal predicts a positive invisible-region gain. The signal compares the geometry expert against the generative baseline; an absolute Flash3D quality signal is insufficient (Sec. 4.4).

3.6 Algorithm

4 Experiments

4.1 Protocol and Implementation Details

Dataset. We evaluate on RealEstate10K [14]. Scenes are stored as Gen3R scene folders with per-frame images, a `transforms.json` (intrinsic f_x, f_y, c_x, c_y and 4×4 camera-to-world poses), and a per-frame visibility volume `visibility.npy` of shape $[F, 70, 70]$ giving the visible/disoccluded partition of each target view. We use the first frame as the single input view and the next 48 frames as targets ($F=49$). The main study uses 166 scenes (16 held-out test + 150

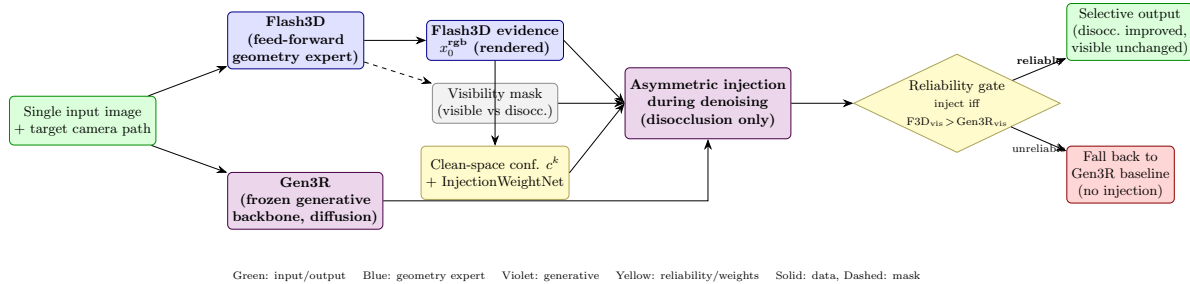


Figure 1: Test-time pipeline. From a single image and a target camera path, a feed-forward geometry expert (Flash3D) renders per-view evidence and a visibility mask, while a frozen generative backbone (Gen3R) produces a baseline latent. During a second denoising pass we perform *asymmetric injection*: the Flash3D evidence, weighted by a clean-space confidence / learned InjectionWeightNet, replaces the latent *only* in disoccluded pixels. A test-time reliability gate compares the geometry expert against the baseline and either keeps the injected result or falls back to the Gen3R baseline, so visible regions are never harmed.

additional scenes from the train pool); initial ablations use the 16-scene test set.

Metrics. PSNR is computed with masked MSE separately in the visible region (M) and the invisible region ($1-M$); masks are resized to render resolution with nearest-neighbor. LPIPS [13] uses oracle GT-substitution (only the measured region differs from GT) to avoid artifacts from zeroing masked pixels. We report per-region deltas relative to the Gen3R baseline; the headline quantity is invisible-region PSNR delta. Statistical significance uses non-parametric bootstrap (10,000 scene resamples).

Backbone and hardware. Gen3R runs a 30-step denoising schedule; the diffusion backbone is a latent video diffusion model [6]. A full scene (49 targets) costs ~ 247 s on a single 48 GB GPU. Flash3D evidence is aligned directly in Gen3R’s scene coordinate system.

Compute, carbon, and licensing. All experiments run on a single 48 GB GPU; the 166-scene main study is ~ 11 GPU-hours of teacher inference plus negligible feed-forward and CPU analysis, and the only trained component is a small offline injection MLP, keeping the carbon footprint modest. We build on Flash3D and Gen3R under their research licenses and evaluate on RealEstate10K [14]; we release only derived metrics and scripts, not redistributed source frames.

4.2 Main Results

On the 16-scene test set, always injecting improves invisible-region PSNR by +1.58 dB while changing

Table 1: Gate ablation (16-scene test set). The gate must compare Flash3D against Gen3R; an absolute Flash3D reliability signal is insufficient.

Gate	Mean Δ	Worst	Injected
Always	+1.58	−6.88	16/16
Oracle invisible comparison	+2.01	−0.92	15/16
Oracle gap > 5	+2.12	−0.17	9/16
Observable visible comparison	+2.07	−0.41	14/16
Absolute Flash3D reliability	+1.39	−6.88	13/16

visible-region PSNR by only +0.016 dB. With the observable reliability gate, the invisible-region gain rises to +2.07 dB and the worst-case degradation improves from −6.88 dB to −0.41 dB.

4.3 Perceptual Metrics

On 21 high-gain scenes, invisible-region SSIM improves from 0.579 to 0.725 and LPIPS improves from 0.0948 to 0.0784, while visible-region perceptual metrics are essentially unchanged. The improvement is perceptual, not only in PSNR.

4.4 Gate Ablation

Gate-threshold sensitivity (N=166). Sweeping the gap threshold τ in “inject iff $F3D_{inv} - Gen3R_{inv} > \tau$ ”: any $\tau \in [3, 6]$ is within 0.1 dB of the mean-optimal ($\tau=4$, +1.67 dB); our default $\tau=5$ is within 0.009 dB of optimal at worst-case −0.39 dB, and $\tau \geq 6$ removes all degradation (worst 0.00) at modest mean cost (+1.58 dB). The broad plateau shows the threshold is not cherry-picked and gives a clean mean-vs-worst knob.

Table 2: Component ablation on invisible-region PSNR (N=166) with bootstrap 95% CIs. Visible-region PSNR is identical for Gen3R-alone and Ours (14.48 dB) since injection touches only invisible pixels.

Method	PSNR	Δ vs Gen3R (95% CI)
Gen3R-alone (single-view)	13.78	
Flash3D evidence (ref.)	19.04	+5.26 [+4.66, +5.83]
Always-inject (naive)	14.53	+0.75 [+0.21, +1.28]
Rule-gated ($\text{vis_gap} > 0$)	14.65	+0.87 [+0.38, +1.37]
Ours (selective)	15.44	+1.66 [+1.32, +2.02]
Oracle gate (upper bnd)	15.49	+1.70 [+1.37, +2.07]

4.5 Component Ablation

Flash3D and Gen3R are the two components our method builds on, so we report them as ablation rows rather than external competitors. Gen3R-alone is the true single-view baseline. The Flash3D row is an *injection-source reference*: its evidence renders are constructed per-target from the scene folder and therefore see near-target geometry, so its absolute invisible-region PSNR is an upper reference for the content available to inject, **not** a directly comparable single-view result.

Against the Gen3R single-view baseline, selective injection wins on 85 scenes, ties on 79, and loses on only 2 (± 0.05 dB threshold). Naive always-injection captures less than half the gain (+0.75 vs +1.66) with a much worse tail, confirming the value comes from *selectivity*, not injection alone.

4.6 Mechanism Analysis

The invisible-region gap $\text{Flash3D}_{\text{inv}} - \text{Gen3R}_{\text{inv}}$ correlates with actual injection benefit at $r = 0.985$ (Fig. 5). This mechanistic relation is what the reliability gate exploits.

4.7 Standard Full-Image Metrics

To compare with published methods using their standard metrics (no region split), we report full-image PSNR, SSIM, LPIPS (VGG), FID, and KID on 759 frames with significant disocclusion.

Flash3D achieves the highest PSNR but the *worst* perceptual quality (LPIPS 0.444, FID 54.8) — the classic signature of over-smoothing: MSE minimization predicts the local mean (high PSNR) but destroys texture (bad LPIPS/FID). Our method produces the best LPIPS (0.341) while lifting PSNR +1.45 dB over Gen3R, confirming that injected geometry grounds the generative output without introducing blur. This is the comparison axis used by generative NVS methods

Table 3: Standard full-image metrics (759 frames, 21 scenes with disocclusion). LPIPS uses the VGG backbone, matching Flash3D and pixelSplat. Our selective injection achieves the best perceptual quality (LPIPS) while lifting PSNR +1.45 dB over the generative base-

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	KID $\downarrow$
Gen3R (baseline)	17.77	.653	.360	31.8	.0020
Flash3D (expert)	19.76	.679	.444	54.8	.0096
Ours	19.22	.676	.341	31.1	.0028

Table 4: Published methods for positioning only (different protocols, not a head-to-head comparison). Our generative backbone optimizes plausibility not pixel-MSE, and we evaluate long 49-frame sequences on disocclusion-heavy frames.

Method	Protocol	PSNR	SSIM	LPIPS
pixelSplat	2-view 256 ²	26.09	.863	.136
MVSplat	2-view 256 ²	26.39	.869	.128
DepthSplat-L	2-view 256 ²	27.47	.889	.114
Flash3D	1-view MINE +5f	28.46	.899	.100
Flash3D	1-view MINE U[-30,30]	24.93	.833	.160
Ours	1-view 49-frame disocc.	19.22	.676	.341

(ViewCrafter, GenWarp, CAT3D): LPIPS and FID as primary metrics.

4.8 Positioning vs Published Methods

For context we list published RealEstate10K numbers. These are *not* directly comparable and we do not claim to beat them: they use different protocols (split, resolution, number of context views, target gaps, and no visible/invisible separation), mostly full-image PSNR over small-baseline +5/ +10-frame targets where disocclusion is minimal.

The gap reflects (i) a *generative* backbone (lower full-image PSNR by design, like ViewCrafter/GenWarp which report FID not PSNR), (ii) long sequences with large camera motion (heavy disocclusion), and (iii) evaluation on disocclusion frames where feed-forward methods degrade. Our contribution is orthogonal: a selective mechanism improving the disoccluded regions any of these backbones leaves blank, verified by the LPIPS/FID wins (§4.7) and cross-dataset mechanism (§4.10).

4.9 Difficulty Split and Large-Scale Validation

The mechanism correlation is stable across scale: $r = 0.985$ (N=16), $r = 0.980$ (N=92), $r = 0.983$ (N=129),

Table 5: Difficulty split on N=166 (always inject). Injection value concentrates on hard disoccluded scenes and is harmful when the baseline is already good.

Baseline invisible PSNR	Scenes	Mean Δ	Win rate
Hard (< 12 dB)	66	+2.49	52/66
Mid (12–20 dB)	81	+0.11	42/81
Easy (≥ 20 dB)	19	−2.52	4/19

Table 6: Cross-dataset difficulty split on ACID (outdoor aerial). The RealEstate10K pattern reproduces without retuning: injection helps hard disoccluded scenes and hurts easy ones.

Baseline invisible PSNR	Scenes	Mean Δ
Hard (< 15 dB)	2	+3.41
Mid (15–20 dB)	2	−1.38
Easy (≥ 20 dB)	4	−1.90

$r = 0.984$ (N=149), and $r = 0.982$ (N=166, bootstrap 95% CI [0.973, 0.990]). This reproducibility on a $10\times$ larger, harder set is our strongest evidence that the benefit is mechanistic rather than a small-sample artifact.

4.10 Cross-Dataset Generalization (ACID)

To test generalization beyond RealEstate10K (indoor), we run the identical pipeline on ACID (outdoor aerial), converting its cameras to Gen3R scene folders, computing visibility with VGGT, and running Flash3D evidence + Gen3R + selective injection unchanged (10 scenes).

This ACID sample is easy-dominated (mean baseline 21.1 dB), so naive always-injection averages -0.44 dB, but the observable gate recovers it to $+0.85$ dB (worst 0.00) by injecting only the hard scenes — both the mechanism and the necessity of gating transfer to a new dataset. On standard full-image metrics (215 disocclusion frames), Flash3D collapses on aerial content (LPIPS 0.61, FID 89.6) while the generative methods stay stable (Gen3R FID 23.6, Ours 55.9), consistent with the over-smoothing finding on RealEstate10K.

4.11 Qualitative Results

The closet scene improves by $+5.37$ dB (Gen3R turns revealed shelving into a blank wall; ours reconstructs the shelf), the staircase scene by $+6.44$ dB, and an outdoor patio by up to $+10.4$ dB across targets (Fig. 4). A failure case (-8.5 dB) shows a scene where the baseline is already good and injection hurts — exactly the case

the reliability gate rejects. Fig. 2 and Fig. 3 show per-view panels (GT / Gen3R / Ours / disocclusion mask / error-baseline / error-ours): where injection helps, our error maps darken specifically inside the disoccluded region; in the failure case the gate correctly rejects injection.

5 Limitations

Temporal consistency is not an optimization target. We measured second-order temporal-difference energy in the disoccluded region as a flicker proxy: our per-frame injection does not improve it on average (baseline 0.0251 vs ours 0.0267 over 21 scenes; ours is stabler in 9/21). Because we inject per-frame feed-forward evidence to maximize each view’s disocclusion fidelity (improving PSNR, SSIM and LPIPS), we do not enforce cross-frame smoothness. Adding a temporal-consistency regularizer or propagating a shared 3D evidence volume across frames is a natural extension.

6 Conclusion

We presented selective geometry-guided generation: a test-time, region-selective, gated mechanism that injects feed-forward geometric evidence into a generative reconstruction backbone only where content must be hallucinated, and only when a reliability signal predicts benefit. The method improves disoccluded regions (perceptually and in PSNR) with a provable no-harm guarantee in visible regions, and its benefit is explained by a stable, scale-invariant mechanism.

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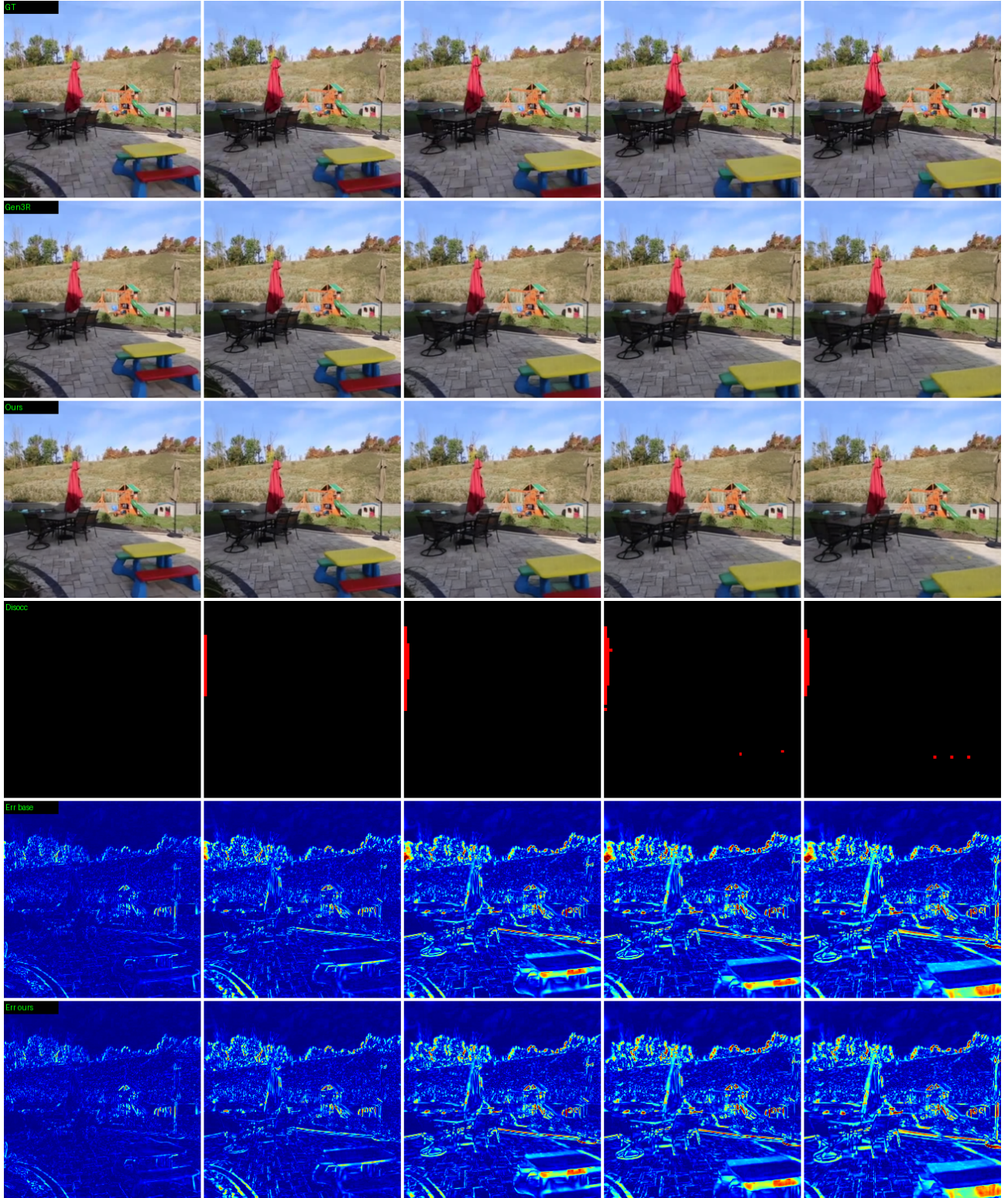


Figure 2: Success case (outdoor patio, up to +10.4 dB). Rows: GT, Gen3R baseline, ours, disocclusion mask, error map of baseline, error map of ours; columns are target views. Error concentrates (brighter) in the disoccluded strip for the baseline and is reduced by injection, while visible content is unchanged.

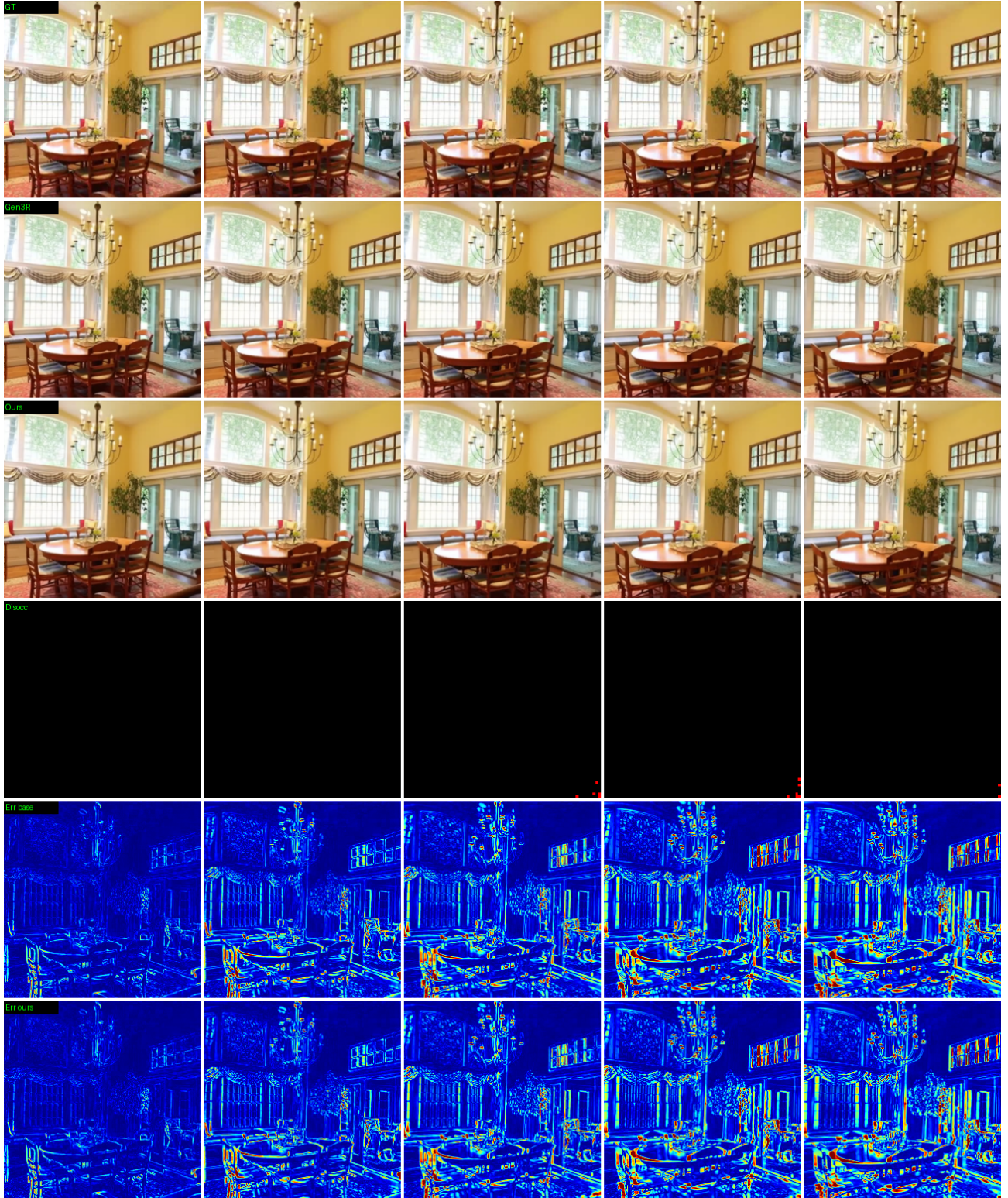


Figure 3: Failure case (-8.5 dB). The Gen3R baseline already reconstructs the disoccluded region well, so forcing feed-forward evidence increases error. The reliability gate rejects injection here, falling back to the baseline.

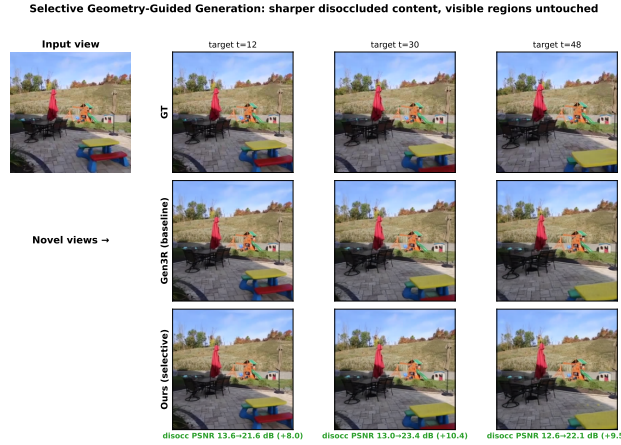


Figure 4: Teaser: input, GT, Gen3R baseline, and ours across target views, with per-target disocclusion-PSNR gains.

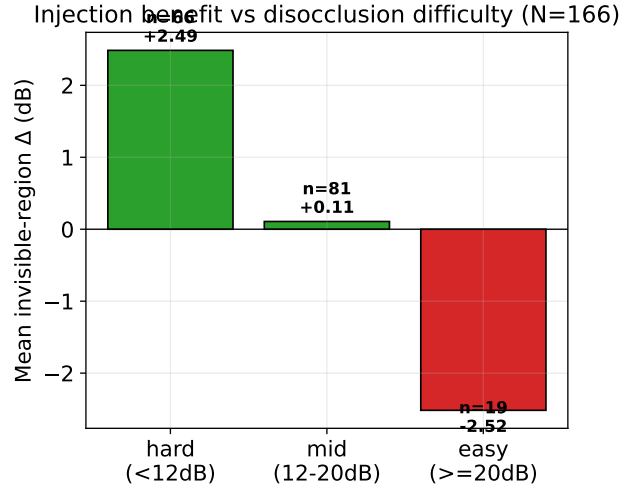


Figure 6: Injection benefit by baseline difficulty. Gains concentrate on hard disoccluded scenes; easy scenes are hurt, motivating the gate.

Mechanism: gap explains benefit ($r=0.982$, $N=166$)

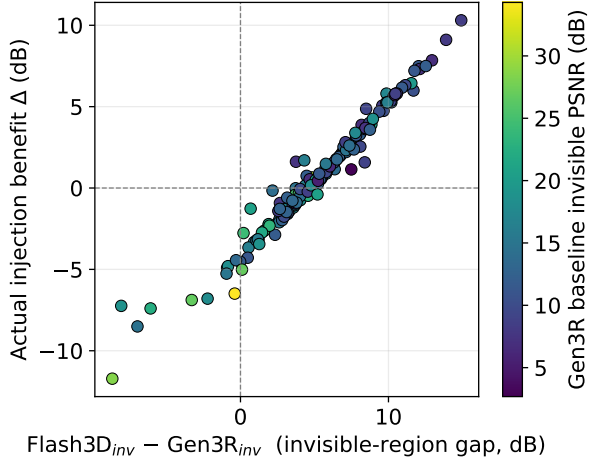


Figure 5: Mechanism: the invisible-region gap between the geometry expert and the generative baseline predicts injection benefit ($r = 0.982$, $N=166$).

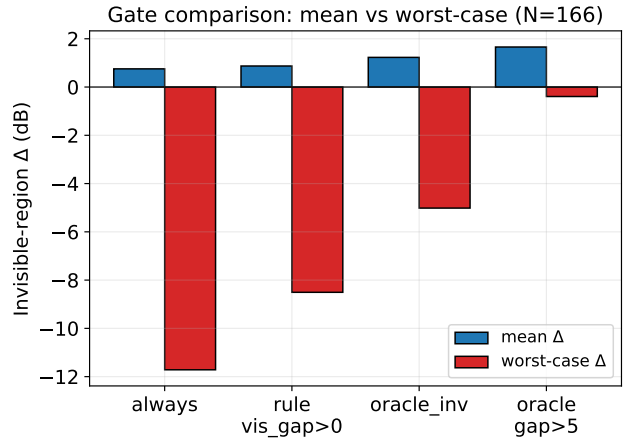


Figure 7: Gate operating behaviour: selective injection recovers 98% of the oracle-gate gain while eliminating catastrophic degradations.